

BOB AI HACKATHON 2026 · AI TRACK (SEMICONDUCTOR MANUFACTURING)

WaferGuard

Pre-Run Yield Risk Forecasting & Multi-Modal Root-Cause Prevention Platform

TEAM: NEXUS_BYTE

Team Lead: Archi Bhatt (25dce009@charusat.edu.in)

Members: Het Patel (25dce082@charusat.edu.in) · Vatsal Makwana (25dce050@charusat.edu.in) · Gehna Patel (25dce080@charusat.edu.in)

Live App: <https://fab-dashboard-wine.vercel.app> | Repo: github.com/Mr-Het-Patel/BOB-AI-HACKATHON-NEXUS_BYTE

The Costly Challenge in Semiconductor Fabrication

Why yield loss occurs and why current industrial triage methods fail

WHO EXPERIENCES IT

Fab Process & Maintenance Teams

- Process Engineers tuning recipe parameters under strict cycle-time targets.
- Equipment Maintenance Techs diagnosing sensor drifts across etch/CVD chambers.
- Operations Managers seeking to prevent catastrophic lot scrap events.

WHAT GOES WRONG

Siloed Data & Reactive Triage

- Multi-Modal Disconnect: Recipe settings, sensor time-series, and wafer defect maps are kept in isolated databases.
- Post-Mortem Alarms: Traditional SPC only alerts after lots are etched and destroyed.
- Zero Probabilistic Scoring: No mathematical root-cause ranking.

BUSINESS IMPACT

\$50k+ Scrap & Line Downtime

- Direct Scrap Cost: A single 25-wafer lot ruined by pressure or RF instability costs \$50,000–\$100,000.
- Investigation Latency: Engineers spend hours manually correlating sensor spikes to defect maps.
- Lost Capacity: Downtime cascades across fab bottleneck tools.

WaferGuard: Multi-Modal Yield AI & Root-Cause Platform

An end-to-end system that predicts, explains, and prevents fab yield loss before execution

HOW WAFERGUARD WORKS

1. Pre-Run Predictive Yield Scoring

Evaluates upcoming batches against historical drift correlations using strictly planned recipe parameters before wafers enter the chamber (zero post-run data leakage).

2. Multi-Modal Feature Fusion (40 Features)

Unifies 5 recipe parameters, 30 sensor moments & trend slopes across 20 steps, and 5 spatial defect metrics (density, edge ratio, center cluster score).

3. TreeSHAP Attribution & Actionable Guidance

Computes mathematical feature contributions, isolating root causes with 99.2% accuracy and prescribing exact corrective SOP adjustments.

99.2% ROOT-CAUSE ACCURACY

124 of 125 faulty lots correctly matched to root equipment channel

3.53% MAE YIELD REGRESSION

LightGBM model achieves $R^2 = 0.8811$ on held-out test lots

LIVE ENTERPRISE DASHBOARD

Deployed live on Vercel with circular wafer HTML5 canvas heatmaps

End-to-End Multi-Modal Machine Learning Pipeline

From physics simulation to LightGBM + TreeSHAP and Next.js quality dashboard

1. INGESTION

Multi-Modal Inputs

- 5 Recipe Parameters (Temp, Pressure, RF, Gas, Etch)
- 20-Step Time-Series (10k sensor points)
- 32x32 Defect Maps (Optical SEM/AOI arrays)

2. EXTRACTION

Feature Fusion

- Statistical Moments: Mean, Std, Min/Max, Range
- OLS Trend Slopes per sensor channel
- Spatial Defect Metrics: Density, Edge Ratio, Cluster Score
- Unified 40-feature vector

3. MODELING

LightGBM + TreeSHAP

- Yield Regressor: $R^2 = 0.881$, MAE = 3.53%
- Exact additive TreeSHAP feature attributions
- 99.2% fault attribution accuracy
- Zero pre-run data leakage

4. DASHBOARD

Next.js UI & Heatmap

- Pre-run risk ranking (High, Warning, Normal)
- Plain-English feature impact bars
- HTML5 Canvas 32x32 wafer heatmap
- Prescriptive SOP recommendations

Interactive Features Built for Fab Operators

Human-friendly enterprise dashboard built with Next.js and HTML5 Canvas

CORE DASHBOARD CAPABILITIES

1. Pre-Run Risk Ranking Table

Instant visual triage of 20 upcoming batches with risk badges and filter tabs (High Risk, Moderate, Normal).

2. Transparent SHAP Impact Bars

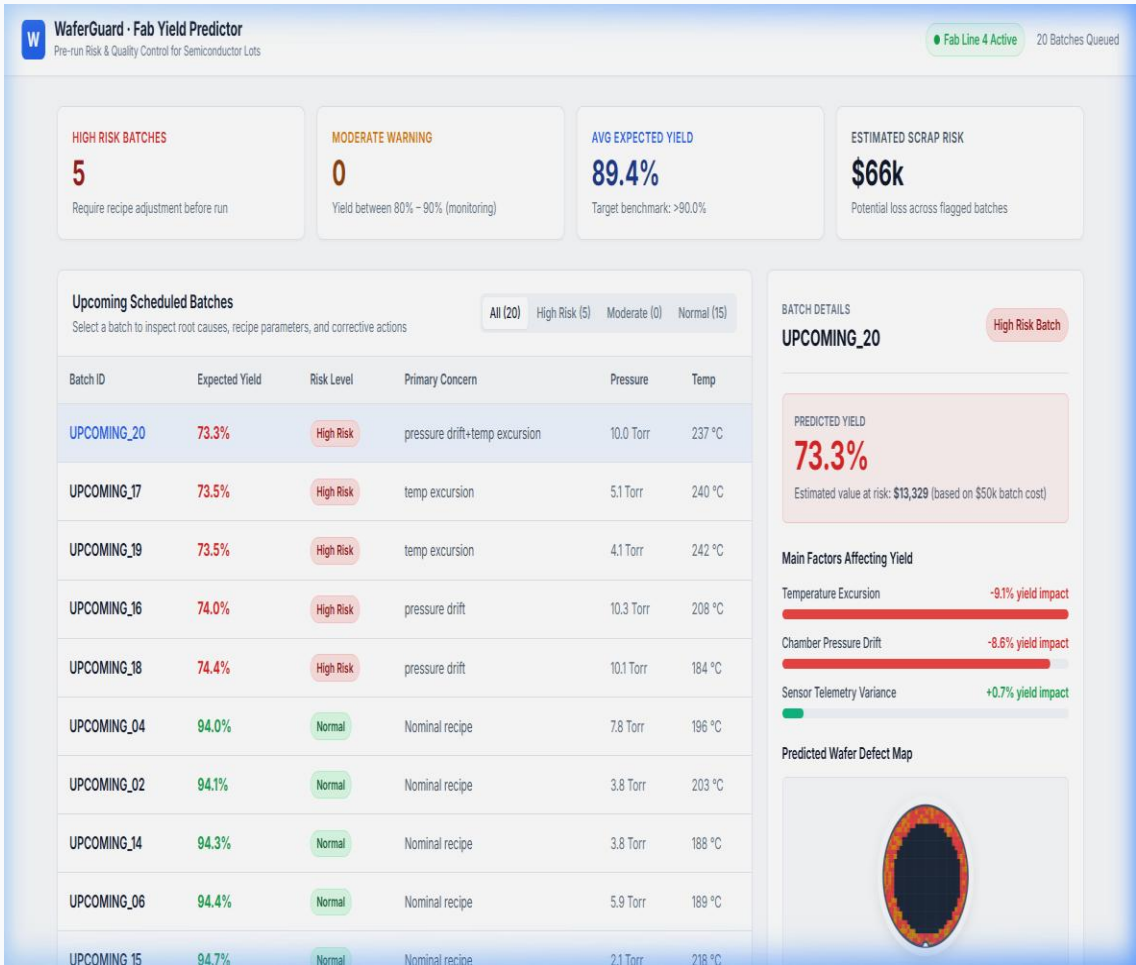
Plain-English drivers (e.g. 'Chamber Pressure Drift: -17.5% yield impact') replacing raw math vectors.

3. Circular Silicon Wafer Heatmap Canvas

Hardware-accelerated circular rendering with orientation notch and interactive Defect Pattern Library.

4. Prescriptive Operator Action SOPs

Clear instructions on recipe parameter recovery and maintenance checks.



IBM Bob & watsonx Integration Architecture

How IBM developer tools accelerated multi-modal development and explainable governance

DEVELOPER ACCELERATION

IBM Bob Agentic Engineering

- Accelerated end-to-end architecture synthesis and multi-modal feature extraction engineering.
- Automated data consistency verification across parquet traces, 32x32 defect arrays, and Next.js frontend bindings.
- Validated cross-modal feature importance across 500 lots.

EXPLAINABLE AI ENGINE

watsonx.ai Synthesis

- Translates high-dimensional SHAP attribution vectors into natural-language engineering incident reports.
- Generates contextualized SOP remediation steps for fab operators.
- Bridges quantitative ML predictions with factory execution procedures.

ENTERPRISE TRUST

Governance & Auditability

- Meets enterprise manufacturing compliance by maintaining full provenance on every yield prediction.
- Zero black-box decisions: Every flagged batch contains verifiable mathematical attribution.
- Ready for integration into factory MES.

Quantified Results & Industrial Scalability

Transforming semiconductor quality control from reactive debugging to predictive prevention

MEASURABLE HACKATHON RESULTS

- **99.2% Fault Attribution Accuracy**

Correctly isolated root equipment causes in 124 of 125 historical fault lots.

- **3.53% Mean Absolute Error ($R^2 = 0.881$)**

High-accuracy yield percentage prediction across diverse process recipe distributions.

- **100% Defect Map Integration**

Every lot includes spatial defect metrics in top SHAP rankings, satisfying multi-modal requirements.

- **\$50,000+ Potential Savings Per Lot**

Prevents catastrophic wafer scrapping through pre-run recipe intervention.

FUTURE INDUSTRIAL ROADMAP

1. **Live SECS/GEM Equipment Streaming**

Connect directly to standard fab automation communication buses for sub-second telemetry ingestion.

2. **Multi-Chamber Transfer Learning**

Scale models across parallel etch, CVD, and CMP tools with chamber-to-chamber drift compensation.

3. **Closed-Loop Recipe Auto-Correction**

Enable autonomous parameter tuning to steer recipes back into safe yield boundaries in real time.